



Lab Book

369.013 Fachvertiefung - Mechatronik und Instrumentierung, SS 2026

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Contents

1 Honor Code

For the assignment I followed the honor code:

- Rule 1: Don't look at solutions or code of other students; everything you submit should be your own work
- Rule 2: Don't share your solution code with others; however, discussing ideas or general strategies is fine and encouraged
- Rule 3: If you use code from somewhere you need to cite the source directly in your code (e.g., StackOverflow). If the source is a generative AI tool (e.g., ChatGPT) provide a link to the specific Chat.

I know that I have full responsibility for what I submit. I made sure the assignment allows the teaching staff to evaluate which competences I have acquired. I am aware that plagiarism is prohibited and may lead to failure of this course.

2 Experiments

The findings in this section are mainly visualized using the image "rubens.jpg" which is referred to as the "source image". Other images used to generate visualizations for this documentation are found in the folder "image". The visualizations themselves are found in the subfolder "documentation/images".

Abbreviations

Tab. 2.1: Abbreviations

σ , sigma	Standard deviation of Gaussian kernel or Gaussian noise
w_k	Kernel width (in pixels)
T_{low}	Low threshold for edge hysteresis
T_{high}	High threshold for edge hysteresis
τ_{low}	Adaptive low threshold for edge hysteresis
τ_{high}	Adaptive high threshold for edge hysteresis

1.1

Defining w_k as a function of σ ensures an accurate representation of the gauss curve in the kernel. w_k is defined to have the following characteristics:

- Odd number of pixels to include the maximum at the center of the kernel.
- w_k of at least $\pm 3\sigma$ around center, so that the kernel covers at least 99 % of the gauss curve.

If w_k is too small, the blur radius is reduced and less blur will occur (see fig. ??). The sharp cutoff will also introduce some distortion, as shown in fig. ??.

If w_k is too large, many near-zero values will be calculated which increases computation time (see tab. ??) but does not contribute to the result (see fig. ??).



Fig. 2.1: Effects of small w_k ($\sigma = 10$). Left: $w_k = 7$ (too small), right: $w_k = 61$ (calculated)



Fig. 2.2: Effects of small kernel width. Left: $\sigma = 2$, $w_k = 13$ (undistorted), right: $\sigma = 10$, $w_k = 7$ (distorted)

Tab. 2.2: Computation time of Gaussian blur, using timeit Python module (100 iterations)

w_k ($\sigma = 10$)	61 px	1000 px
computation time	1075 ms	10 263 ms

1.2

Blurring (using a linear filter) smooths out the pixel intensities without changing their mean value. This is because the linear filter is a moving weighted average of each pixel neighborhood where the sum of coefficients is always 1. Fig. ?? illustrates this principle for a single row of pixels of a given image.

2.1

T_{low} determines the lowest possible gradient for a pixel to be considered part of an edge. A large T_{low} might therefore introduce holes in otherwise



Fig. 2.3: Choosing a larger than necessary w_k has no practical effect. Left: $w_k = 61$, right: $w_k = 1000$; $\sigma = 10$

continuous edge loops.

T_{high} determines the lowest possible gradient for a continuous edge to be considered valid. A large T_{high} removes consistently smooth edges (smooth = small gradient) while keeping other edges with at least one sharp section intact.

Fig. ?? shows the effect of varying T_{low} . Fig. ?? shows the effect of varying T_{high} . See tab. ?? for a summary of the effects.

Tab. 2.3: Effects of thresholds

	smaller value	larger value
T_{low}	more edges overall	fewer edges overall
T_{high}	more edge segments	fewer edge segments

2.2

Since T_{low} and T_{high} limit gradients, images with different contrast (due to different environments and lighting) will yield different results for the

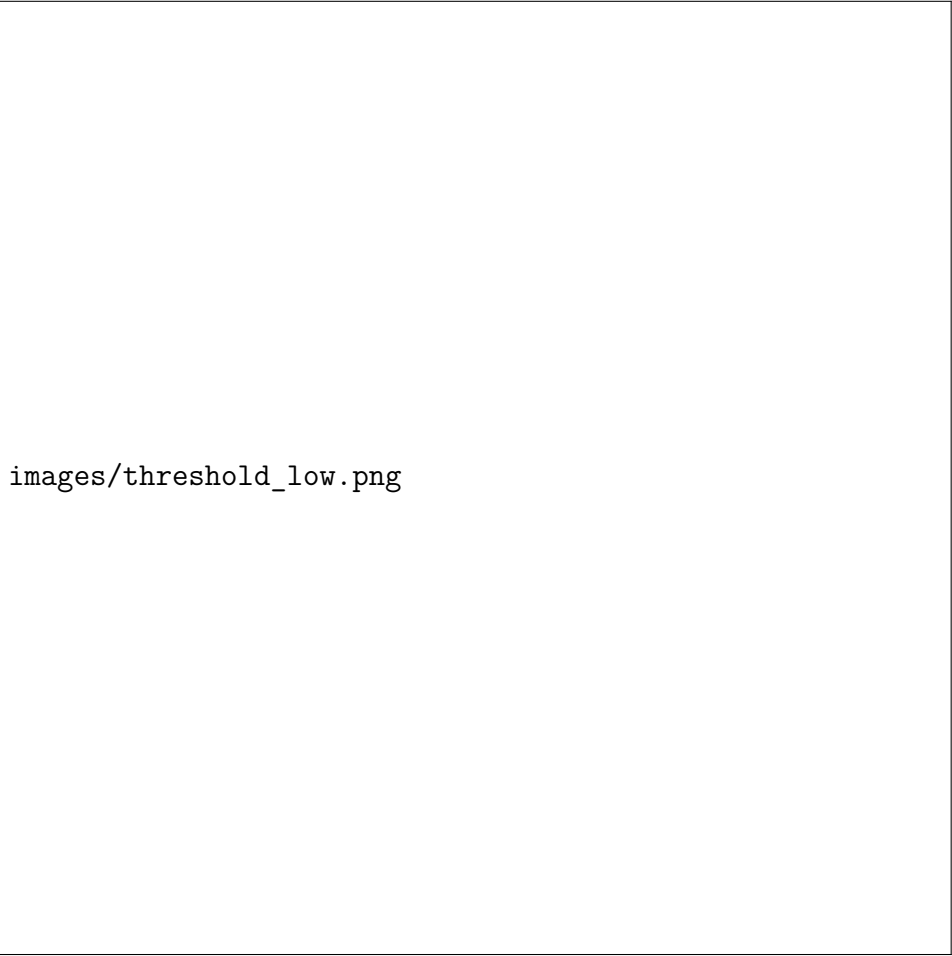


Fig. 2.4: Averaging effect of blur using pixel row 100 of the source image: Pixel intensities with no blur (left) and with Gaussian blur ($\sigma = 5$, $w_k = 31$) (right)

same thresholds.

Adaptive thresholding aims at achieving a largely consistent amount of edges across different environments. However, fixed thresholding might be more suitable in a stable environment where the amount of detected edges is important and may vary greatly. For example, we might wish to estimate the number of objects on a table. The more objects, the more edges should be detected.

See fig. ?? for a comparison of both methods. τ_{low} and τ_{high} have been chosen to achieve a similar amount of edges in the top image. We can see that in the bottom image, the adaptive method excludes some objects



images/threshold_low.png

Fig. 2.5: Edge hysteresis with varying T_{low} ($T_{\text{high}} = 0$)

because of the overall edge limit given by τ_{high} , whereas the fixed method detects objects more reliably in both images.

3.1

Since noise has very high frequency components, it will introduce sharp gradients throughout the whole image which might be detected as edges. See fig. ?? for an illustration of this effect.

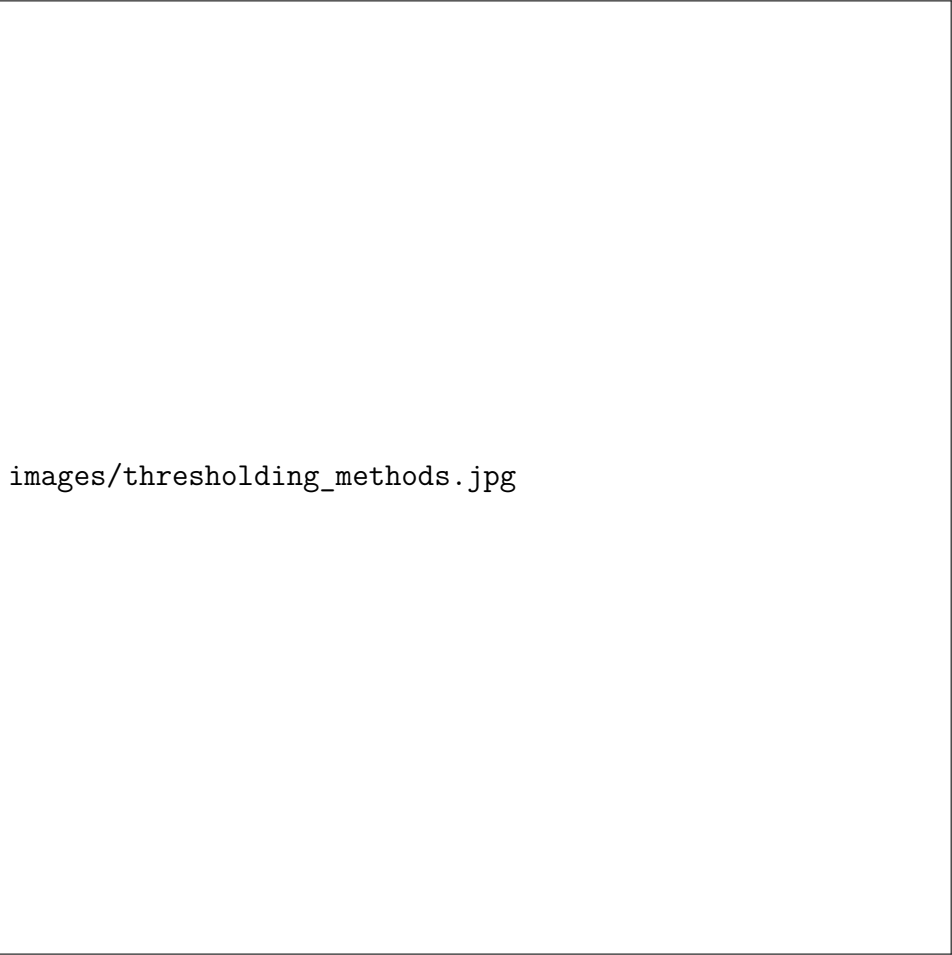


Fig. 2.6: Edge hysteresis with varying T_{high} ($T_{\text{low}} = 0$)

3.2

Because blurring works as a lowpass filter, applying a wide gauss filter (large σ) will get rid of high frequency noise and mitigate sharp gradients introduced by it. We can see the lowpass characteristic in fig. ???. The soft gradients introduced by the low frequency components of the noise can then be sorted out by increasing the hysteresis thresholds.

It is therefore possible to set the parameters such that edges can be detected *as if almost no noise was present (!)*, even in highly noisy images. However, finer details will be lost due to the blurring (see fig. ??).

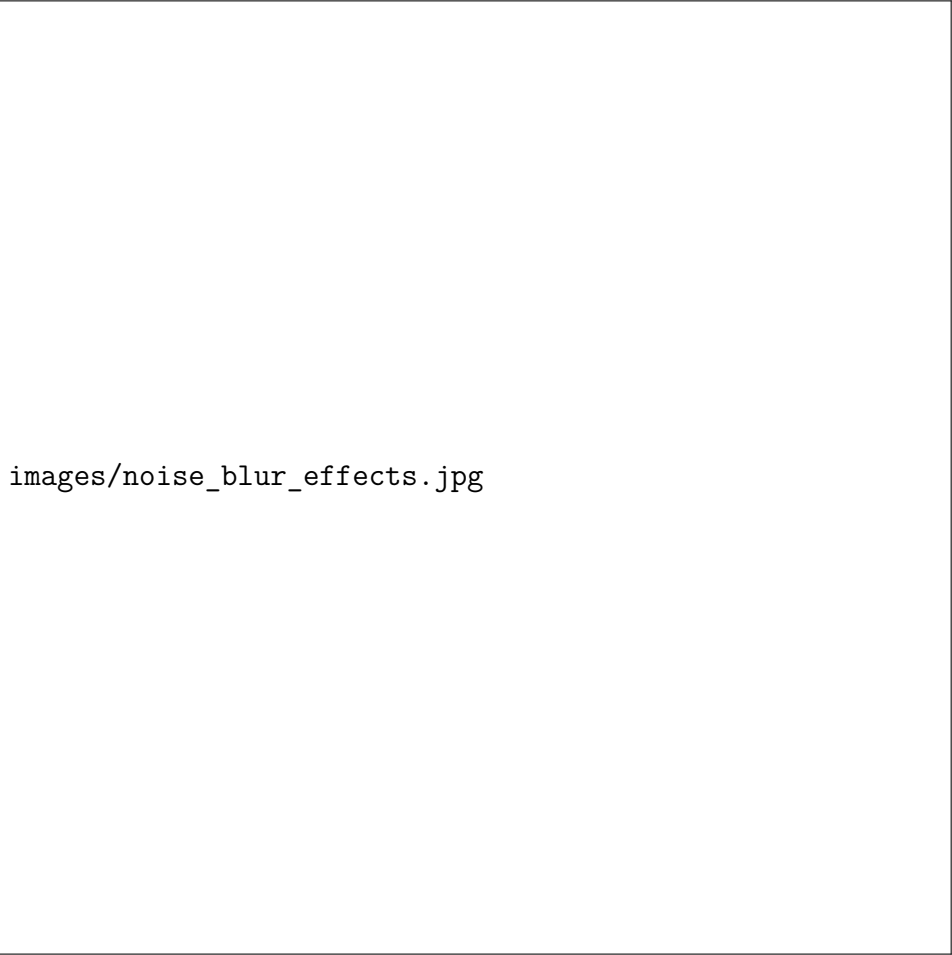


`images/thresholding_methods.jpg`

Fig. 2.7: Comparison of fixed and adaptive thresholding methods using images “table_object.jpg” and “table_objects.jpg”. Note that some objects like the glasses or the compass are not detected by the adaptive method.



Fig. 2.8: Effects of noise on edge detection without blur.



images/noise_blur_effects.jpg

Fig. 2.9: Left: Source image with noise, right: Canny edges with various blur sigmas (no edge hysteresis applied). Note how increasing σ reduces the frequency of detected edges.

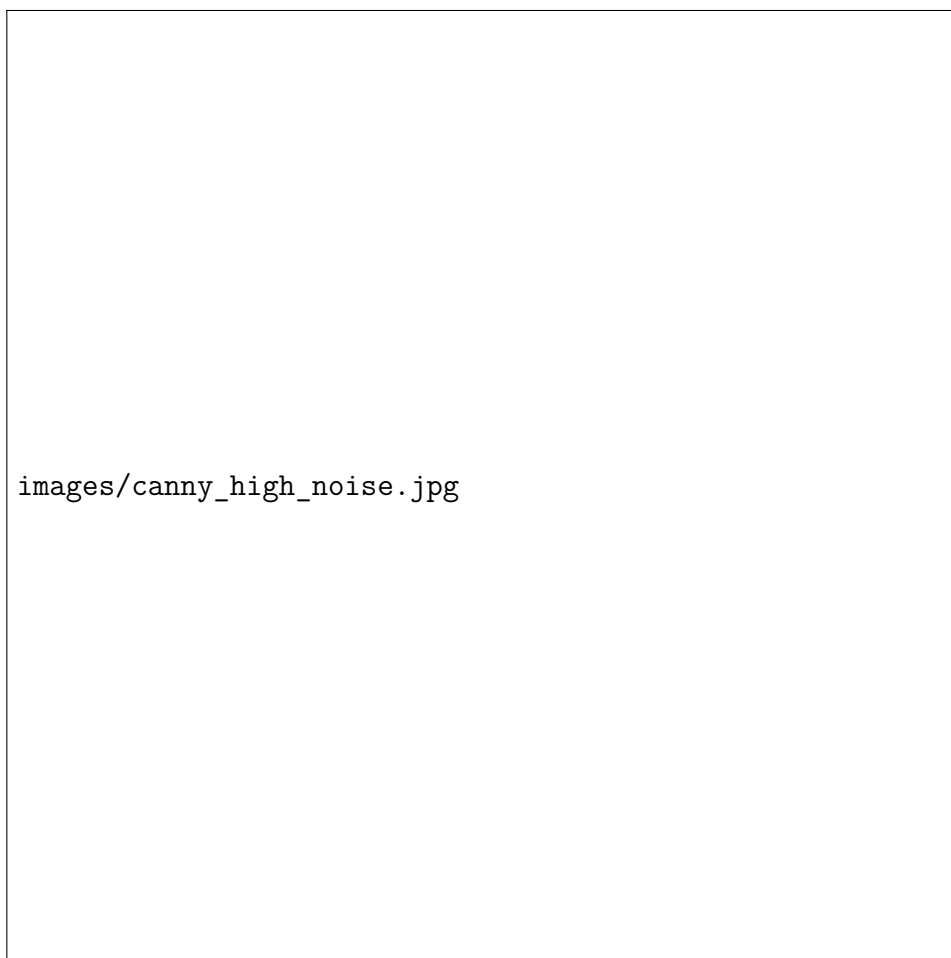


Fig. 2.10: Top: Source image, bottom: image with high noise ($\sigma = 0.5$). Blur $\sigma = 10$, $T_{\text{low}} = 0.1$, $T_{\text{high}} = 0.2$ for both images.